

EXP NO: 3
DATE:

Google App Engine SDK and Web App Development

Aim:

To install the Google App Engine SDK and develop a Hello World / simple web application using Python.

Pre-Requisites:

- Windows 10 Host Machine
- Oracle VM VirtualBox (Open Source)
- Ubuntu Desktop ISO image (Open Source)

Procedure:

Step 1: Install Google Cloud CLI & Python Ensure

Python 3 is installed on the system.

```
ubuntu@ubuntu:~$ sudo apt install python3
python3 is already the newest version (3.14.3-0ubuntu2).
python3 set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 3
```

Download and install the Google Cloud CLI (contains the modern App Engine SDK) from cloud.google.com/sdk/docs/install:

```
curl -O
https://dl.google.com/dl/cloudsdk/channels/rapid/downloads/googlecloud-
cli-linux-x86_64.tar.gz
tar -xf google-cloud-cli-linux-x86_64.tar.gz ./google-cloud-sdk/install.sh
```

Step 2: Initialize GCloud & Install Components

```
gcloud init
```

Sign in to your Google account when prompted.

```

lakshiya@lakshiya-VirtualBox:~$ gcloud init
Welcome! This command will take you through the configuration of gcloud.

Your current configuration has been set to: [default]

You can skip diagnostics next time by using the following flag:
  gcloud init --skip-diagnostics

Network diagnostic detects and fixes local network connection issues.
Checking network connection...done.
Reachability Check passed.
Network diagnostic passed (1/1 checks passed).

You must sign in to continue. Would you like to sign in (selecting "Y" will open your browser to the sign-in page where you
complete authentication)
(Y/n)? y

Your browser has been opened to visit:
  https://accounts.google.com/o/oauth2/auth?response_type=code&client_id=32555940559.apps.googleusercontent.com&redirect_uri=http%3A%2F%2Flocalhost%
3A8885%2F&scope=openid+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcloud-platform+https%3A%2F%
2Fwww.googleapis.com%2Fauth%2Fappengine.admin+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fsqlservice.login+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcorpora
https%3A%2F%2Fwww.googleapis.com%2Fauth%2Faccounts.reauth&state=LbPLNpuKjLz8QztlCEMLAkwzAatnp&access_type=offline&code_challenge=V5iZKitWJ1WL_Kc
m0p8tJ8I53x59QPCuFP_IFLE&code_challenge_method=S256

```

gcloud components install app-engine-python

```

lakshiya@lakshiya-VirtualBox:~$ gcloud components install app-engine-python

Your current Google Cloud CLI version is: 574.0.0
Installing components from version: 574.0.0

Performing in place update...

These components will be installed.



| Name                                    | Version | Size     |
|-----------------------------------------|---------|----------|
| Cloud Datastore Emulator                | 2.3.1   | 36.2 MiB |
| gRPC Python library (Platform Specific) | 1.70.0  | 5.6 MiB  |
| gcloud app Python Extensions            | 1.9.118 | 3.8 MiB  |



For the latest full release notes, please visit:
https://cloud.google.com/sdk/release\_notes

Once started, canceling this operation may leave your SDK
installation in an inconsistent state.

Do you want to continue (Y/n)? Y

Performing in place update...

Performing post processing steps...done.
Update done!

Download: Cloud Datastore Emulator
Download: gRPC Python library
Download: gRPC Python library (Platform Specific)
Download: gcloud app Python Extensions
Installing: Cloud Datastore Emulator
Installing: gRPC Python library
Installing: gRPC Python library (Platform Specific)
Installing: gcloud app Python Extensions

```

Step 3: Create the App Directory

```

mkdir gae-hello-world cd
gae-hello-world

```

Step 4: Create Application Code (main.py)

```

from flask import Flask app =
Flask(__name__)

```

```

@app.route("/") def hello():      return "Hello
World from Google App Engine!"
    if __name__ == "__main__":
app.run(host="127.0.0.1", port=8080, debug=True Step

```

5: Create Configuration File (app.yaml)

```

runtime: python39
handlers: - url:

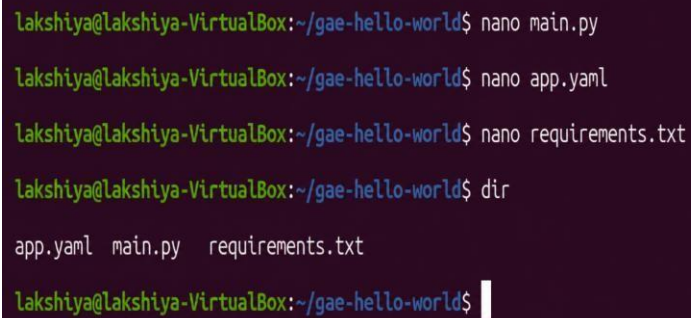
```

```
/* script:  
auto
```

Step 6: Create requirements.txt Flask==2.0.1

Output:

Files created in project directory (main.py, app.yaml, requirements.txt):

A terminal window with a dark purple background and light green text. The prompt is 'lakshiya@lakshiya-VirtualBox:~/gae-hello-world\$'. The user enters 'nano main.py', 'nano app.yaml', and 'nano requirements.txt' on separate lines. After the third command, the prompt returns. The user then enters 'dir', and the output 'app.yaml main.py requirements.txt' is displayed on the next line. The prompt returns again.

```
lakshiya@lakshiya-VirtualBox:~/gae-hello-world$ nano main.py  
lakshiya@lakshiya-VirtualBox:~/gae-hello-world$ nano app.yaml  
lakshiya@lakshiya-VirtualBox:~/gae-hello-world$ nano requirements.txt  
lakshiya@lakshiya-VirtualBox:~/gae-hello-world$  
app.yaml main.py requirements.txt  
lakshiya@lakshiya-VirtualBox:~/gae-hello-world$
```

Result:

Thus, the Google App Engine SDK was installed and a simple Hello World web application was developed using Python and Flask.